Clinica Chimica Acta 565 (2025) 120022

Contents lists available at ScienceDirect

Clinica Chimica Acta

journal homepage: www.elsevier.com/locate/cca

Research Paper

UPLC-Q-TOF/MS-based urine metabolomics for the diagnosis and staging of bladder cancer

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ARTICLE INFO

Keywords:
Bladder cancer
Non-targeted metabolomics
Binary logistic regression
Biomarker panel

ABSTRACT

Background: Bladder cancer (BC) is a common malignant tumour of the urinary system. Currently, the gold standard for diagnosing BC is cystoscopy, but it is an invasive examination that can lead to a certain psychological burden on the patient. In this study, we aimed to identify non-invasive potential metabolic biomarkers that could improve the diagnostic accuracy of bladder cancer.

Methods: Urine from 30 healthy people and 50 BC patients, including 40 non-muscle-invasive bladder cancer (NMIBC) patients and 10 muscle-invasive bladder cancer (MIBC) patients, were analyzed by liquid chromatography coupled with mass spectrometry to identify potential diagnostic metabolites. Binary Logistic regression was used to construct biomarker panels. Correlation analysis and construction of compound-reaction-enzyme-gene network were also performed to explore the possible mechanisms of BC development.

Results: Twenty-six metabolites were identified for differentiating BC patients from healthy controls, and eight metabolites were identified for differentiating NMIBC patients form MIBC patients. The biomarker panel consisting of urate, 4-Androstene-3α, 17β-diol and 3-Indoxyl sulfate can distinguish well between BC patients and healthy controls, with an area under the ROC curve (AUC) value of 0.983. And the biomarker panel consisting of L-Octanoylcarnitine, γ-Glutamylleucine, and heptanoylcarnitine for distinguishing NMIBC patients from MIBC patients had an AUC value of 0.941.

Conclusions: The diagnostic capability of the biomarker panels are superior to that of any single potential biomarker. This panel significantly benefits bladder cancer diagnostics and reveals insight into bladder cancer pathogenesis.

1. Introduction

Bladder cancer is a common malignant tumour of the urinary system, the incidence of which is steadily increasing globally, with up to approximately 550,000 new cases and 200,000 deaths each year [1]. Based on the developmental process, five stages of bladder cancer were classified, including T0 (Ta: noninvasive papillary carcinomas and Tis: tumor in situ), T1 (Tumor invades the bladder mucosa), T2 (Tumor invades the bladder inner muscle), T3 (Tumor invades the bladder outer muscle) and T4 (Tumor metastasis). According to the degree of spread, T0 and T1 stages belong to non-muscle-invasive bladder cancer (NMIBC), T2 stage and above belong to muscle-invasive bladder cancer (MIBC) [2].

Early diagnosis is the key to improving the survival rate of bladder cancer. Currently, cystoscopy, urine cytological examination and imaging examination are the main methods to diagnose bladder cancer [3]. Cystoscopy has always been the gold standard for diagnosing bladder cancer, but it is an invasive examination that can lead to a certain psychological burden on the patient and expensive [2]. In addition, it can cause urethral injuries, urinary tract infections and other

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https://doi.org/10.1016/j.cca.2024.120022
Received 14 May 2024; Received in revised form 10 October 2024; Accepted 25 October 2024
Available online 28 October 2024
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complications. Urine cytology, as a non-invasive examination, has a high specificity (86–99%), but its sensitivity is low (48–70%) and correlates with the type of bladder cancer [4]. The limitation confines its use in low-grade bladder cancer. Therefore, it is urgent to find a non-invasive and inexpensive diagnostic method which have high specificity and sensitivity for the early diagnosis of bladder cancer.

Metabolomics is a study of high-throughput analyses of small molecules (<1500 Da) that can be used to understand interactions within biological systems and the effects of disease on metabolic processes in biological systems [5]. Nowadays, metabolomics has been used to discover biomarkers for diagnosing diseases, searching for new therapeutic targets, and explaining the mechanism of drugs. The molecular pathogenesis of cancer is closely related to metabolism [6,7]. In recent years, more and more researches showed that metabolomics can be used for discovering potential biomarker to diagnose cancer. Han et al. analysed tissue and serum samples from patients with hepatocellular carcinoma by metabolomics and identified four potential biomarkers (chenodeoxycholic acid, LPC20:5, succinyladenosine, uridine) as a biomarker panel for the diagnosis of hepatocellular carcinoma, with 96.7% accuracy [8]. Jajin et al. analysed plasma from patients with medullary thyroid carcinoma (MTC) through gas chromatography-mass spectrometry-based untargeted metabolomics, and find that linoleic acid (AUC=0.935), linolenic acid (AUC=0.920), and leucine (AUC=0.948) could discriminate MTC from healthy groups [9].

Currently, many studies on bladder cancer metabolomics are based on NMR, liquid and gas chromatography coupled with mass spectrometry [10]. Wang et al. analyzed metabolites in the urine of 29 bladder cancer patients and 15 healthy controls using UPLC-MS and identified a total of 19 potential bladder cancer biomarkers suggesting that metabolomics could be a non-invasive early detection method for bladder cancer [11]. Similarly, urine profiling was performed by Jin et al. on a much larger group of 138 BCE patients and 121 healthy controls, indicating 12 potential BC biomarkers [12]. Metabolomics can also be used to differentiate between different stages of bladder cancer. Tan et al. obtained three biomarkers for differentiating low-grade bladder cancer from high-grade bladder cancer by bladder cancer serum metabolome [13]. Oto et al. used mass spectrometry-based metabolomics to discover novel urine metabolomic profiles to diagnose and stage NMIBC and MIBC patients on much larger sample sets including 198 BCE samples and 98 healthy volunteers, with external validation applied, and they find that p-cresol glucuronide (AUC=0.790), p-coumaric acid (AUC=0.803) and spermine (AUC=0.882) can be used to diagnose BC, NMIBC and MIBC respectively, with specificity and sensitivity in the range of 70%–80% [14]. Nizioł et al. analyzed 200 cases of human urine to reveal the biomarkers that distinguish BC patients from healthy individuals, with external validation applied, and discussed more detailed delineation of BC staging, age and gender [15]. Despite many studies on bladder cancer metabolomics, there are no biomarkers that have good diagnostic accuracy, sensitivity and specificity at the same time to diagnose bladder cancer and its staging, and none of them are used in the clinic. In addition, there are very few reports on the analysis of the relationship between BC, stage, age, and gender.

In this work, 80 human urine samples, including 50 bladder cancer patients and 30 healthy controls, were profiled using high-performance liquid chromatography coupled with high-resolution mass spectrometry (HPLC-HRMS), which aim to identify the novel potential biomarkers of bladder cancer with good diagnostic accuracy, sensitivity and specificity, and explored the relationship among BC, stage, age, and gender. Then validate the potential biomarkers through receiver operating characteristic (ROC) curve analysis, construct a compound-reaction-enzyme-gene network to trace the genes associated with differential metabolites. The results of the study show that the biomarker panel we obtained can accurately diagnose bladder cancer and its staging, which the AUC value, sensitivity and specificity were greater than 0.9. And we found that bladder cancer development may be associated with disturbances in energy metabolism and amino acid metabolism.

2. Materials and methods

2.1. Chemicals and reagents

Acetonitrile and methanol of HPLC grade were purchased from Merck (Darmstadt, Germany). Formic acid and ammonium acetate of LC-MS grade were purchased from Merck (Waltham, USA). Ultrapure water was prepared by the Milli-Q Advantage A10 system (Millipore, Bedford, MA, USA).

2.2. Participants

The urine samples in this study were collected from the participants (>18 years old), including 50 bladder cancer patients and 30 healthy individuals, which recruited at the first affiliated hospital of Sun Yat-Sen University from July 2022 to June 2023. These bladder cancer patients were diagnosed by the attending medical team according to the pathologically diagnosed. Based on the pathological diagnosis, they were classified into MIBC and NMIBC. The inclusion criteria for the healthy controls in this study were as follows: (1) No chronic diseases such as other malignant tumors, hypertension, diabetes, etc.; (2) No other urinary tract disease; (3) Age range 18–80; (4) All subjects had given informed consent to be included in this study before participation. Table 1 summarizes the basic participants information. Statistical analyses were performed using t-tests for continuous variables (age, BMI and GFR) and chi-square tests for categorical variables (gender, smoking history and drinking history). This study was approved by the Clinical Research of the First Afiliated Hospital of Sun Yat-Sen University (IRB no. [2024]240) and institutional review board of Guangdong Provincial Health Bureau and Sun Yat-sen University. All subjects had given informed consent to be included in this study before participation.

2.3. Urine sample collection and preparation

Collected urine samples were centrifuged at 12,000 rpm for 10 min at 4 °C to remove particle contaminants, then the supernatant was collected and transferred to tubes and stored at −80 °C until processing. Before analysis, frozen urine samples were kept at 4 °C to thaw, then vortexed for 20 sec, then mix and dilute with an equal volume of acetonitrile, centrifuged at 12,000 rpm for 10 min, and the supernatant was transferred to new vials for analysis. In order to evaluate the stability of the analytical method during the analysis, quality control (QC) samples were inserted for analysis prior to sample analysis, between every eight samples and before the end of analysis. The QC samples were prepared by a mixture of bladder cancer samples and healthy controls samples with the same volumes.

2.4. UPLC-Q-TOF/MS analysis

LCMS-9030 (Shimadzu, Japan) were used to perform UPLC-QTOF/MS analysis of urine samples. Chromatographic separation was

Table 1
Characteristics of enrolled patients.

Group		BC patients	Healthy controls	P value
Number of subjects		50	30	–
Stage	NMIBC	40	–	–
	MIBC	10		
Gender (Male, %)		68.00%	83.33%	0.131
Age, years (mean ± SD)		64.92 ± 12.73	52.17 ± 11.93	0.0001
BMI (mean ± SD, kg/m^2)		21.90 ± 4.03	22.17 ± 3.54	0.766
Smoking (Yes)		18.00%	6.67%	0.154
Alcoholic (Yes)		8.70%	6.67%	0.826
GFR (mean ± SD, ml/min)		75.72 ± 22.34	77.40 ± 22.95	0.747

BC: bladder cancer; NMIBC: non-muscle-invasive bladder cancer; MIBC: muscle-invasive bladder cancer.

performed on a Waters ACQUITY UPLC HSS T3 column (2.1 mm × 100 mm, 1.8 μm) at 35 °C. A linear gradient elution of H_2O (with 0.1% formic acid) (A) and acetonitrile (with 0.1% formic acid) (B) was used: from 0 to 2 min, 3.0% B; from 2.0 to 17.0 min, 3.0–35.0% B; from 17.0 to 22.0 min, 35.0–95.0% B; from 22.0 to 25.0 min, maintained 3.0% B. The flow rate was set at 0.3 mL/min. The volume of the injection for each sample was 2 μL.

The mass data were acquired under following conditions: The temperatures of the interface, heat block, and desolation line were set at 300 °C, 400 °C, and 250 °C, respectively. The drying and nebulizer gases were set at 10 L/min and 1 L/min. The data acquisition processes were all calibrated in real time with 400 mg/L sodium iodide solution for the mass axis. The acquisition of the data was operated by Labsolution software (Shimadzu, Japan).

2.5. Data processing and statistical analysis

Data pre-processing, such as peak picking and alignment, were performed in MS-DIAL software version 4.38. The parameters used were MS1 and MS2 tolerances of 0.005 and 0.05, respectively; minimum peak height of 2000; mass slice width of 0.05 Da; linear-weighted moving average as the smoothing method using 3 scans and peak width of 5 scans; sigma window value for deconvolution of 0.5; 0.1 min and 0.005 Da tolerance for peak alignment. The QC sample in the middle of the batch was selected as the peak alignment file. The data aligned in MS-DIAL was uploaded to the Metaboanalyst platform (https://www.metaboanalyst.ca/) to filter the features which RSDs are > 30% in QC samples, remove features with > 20% missing values, replace the missing values with the mean value, and normalize by sum.

Then principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) were performed using SIMCA 14.1 after pareto scaling. The variable importance in projection (VIP) score, calculated based on the OPLS-DA model, was applied to screen the metabolites that can best distinguish the two groups. Student's *t*-test and Fold change analysis were then performed using GraphPad Prism 8.3.0 and Metaboanalyst and finally variables with VIP > 1 and $p < 0.05$ were defined as differential metabolites. To ensure the validity of the student's *t*-test, we verified the test assumptions, including normality of distribution and homogeneity of variance between groups. Student's t-tests were performed for data that conformed to normal distribution and homogeneity of variance. Welch's t-tests were performed for data that conformed to normal distribution and heterogeneity of variance, and non-parametric tests (Mann-Whitney tests) were performed for data that did not conform to normal distribution. For multiple comparisons, we use Bonferroni method to control for the risk of type I errors. AVOVA tests were performed for data that conformed to normal distribution and homogeneity of variance. Welch AVOVA tests were performed for data that conformed to normal distribution and heterogeneity of variance, and used Dunnett T3 method to correct for multiple comparisons. Non-parametric tests (Kruskal-Wallis tests) were performed for data that did not conform to normal distribution using Dunn's test to correct. Differential metabolites were further identified by utilizing MS-DIAL, Pubchem (https://pubchem.ncbi.nlm.nih.gov/) and Human Metabolome Database (HMDB) (http://www.hmdb.ca/) with the information about the accurate molecular mass, and mass spectrum [16]. Metabolic pathway analysis was performed in Metaboanalyst 5.0 using the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway library for Homo sapiens to uncover the significantly disturbed pathways affected by bladder cancer. The biomarker panel was further defined by building a binary logistic regression model (Forward: LR), carried out by SPSS 19.0 (IBM, U.K.), and evaluated its diagnostic capabilities by receiver operating characteristic (ROC) curve analysis, which was performed by Metaboanalyst 5.0 [17]. Moreover, we evaluated the correlation of differential metabolites, stage, age and gender, through the Pearson correlation coefficient r value [18]. Correlation analysis was performed in Hiplot Pro (https://hiplot.com.cn/). Finally, to visualise genes associated with differential metabolite, we constructed compound-reaction-enzyme-gene network through MetScape [19].

3. Results

3.1. Demographic and clinical information

A total of 80 participants were enrolled including 30 healthy controls and 50 cases of bladder cancer. Table 1 summarizes the basic participants information. Statistical analyses were performed using t-tests for continuous variables (age, BMI and GFR) and chi-square tests for categorical variables (gender, smoking history and drinking history). There were no significant differences in gender ratio, BMI, GFR, smoking history and drinking history of the participants between the healthy controls and disease groups ($p > 0.05$). Although there was a significant difference in age between BC patients and healthy controls, we grouped the age of the sample by quartiles for PCA analysis. The results showed that there was no trend of separation between groups of different ages, suggesting that the effect of age on metabolites was small (Supplemental Fig. 1).

3.2. Distinguishing between bladder cancer and healthy controls urine samples

In order to evaluate the stability of the analytical method during the batch, real samples together with QC samples were analysed periodically. As shown in Supplemental Fig. 3, all of the QC samples are within twice the standard deviation (SD), which indicates that the analytical method is stable.

Multivariate statistical analyses were performed using SIMCA 14.1. In the PCA scores, there was a clear trend of separation between the bladder cancer group and the healthy controls (Supplemental Fig. 4). As shown in Fig. 1A-B, OPLS-DA mode presents good fitting and prediction ability with $R^2Y=0.947$ and $Q^2=0.857$ in positive mode and $R^2Y=0.943$ and $Q^2=0.861$ in negative mode, BC patients are clearly distinguished from healthy controls. Furthermore, the models were confirmed by the 200 permutation test indicating that they were without overfitting (Supplemental Fig. 5) and can be used in the next step of the analysis. Then, VIP values based on the OPLS-DA model were used to screen for differential metabolites. VIP > 1 as one of the indicators for screening differential metabolites.

Student's *t*-test and Fold change analysis were then performed using GraphPad Prism 8.3.0 and Metaboanalyst. To ensure the validity of the student's *t*-test, we verified the test assumptions, including normality of distribution and homogeneity of variance between groups. Student's t-tests were performed for data that conformed to normal distribution and homogeneity of variance. Welch's t-tests were performed for data that conformed to normal distribution and heterogeneity of variance, and non-parametric tests (Mann-Whitney tests) were performed for data that did not conform to normal distribution. The volcano plots (Fig. 1C-D) are generated to show the fold change (FC > 0) and p-value ($p < 0.05$) indicators. Finally, variables with VIP > 1 and $p < 0.05$ were defined as differential metabolites.

The differential metabolites obtained from the above screening were identified by MS-DIAL, HMDB and Pubchem. A total of 26 differential metabolites were identified after excluding duplicates and external substances (Fig. 2, Supplemental Table 1).

3.3. Distinguishing between NMIBC and MIBC urine samples

Likewise, multivariate statistical analyses were performed using SIMCA 14.1. In the PCA scores, there was a trend of separation between NMIBC patients and MIBC patients (Supplemental Fig. 4). As shown in Fig. 1E-F, NMIBC patients and MIBC patients have a tendency to separate in PLS-DA model, which presented good fitting and prediction

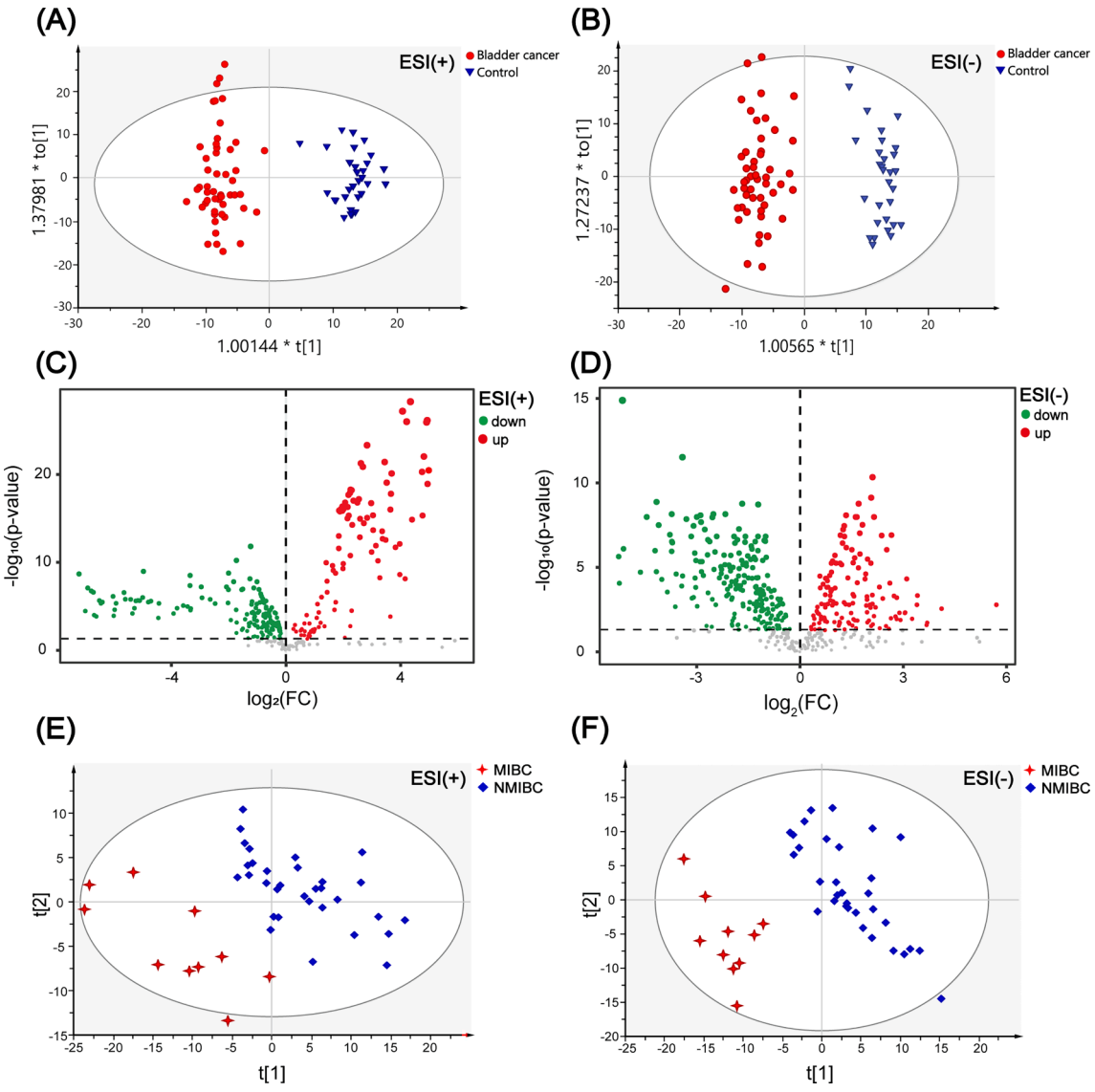


Fig. 1. Analysis of metabolomics data from BC patients and healthy controls. $n = 30$ in the healthy controls group, $n = 50$ in the bladder cancer group (including 40 NMIBC patients and 10 MIBC patients). (A) and (B) OPLS-DA scores plots of BC patients and healthy controls. (C) and (D) Volcano plots of BC patients and healthy controls. (E) and (F) PLS-DA scores plots of NMIBC patients and MIBC patients.

ability with R^2Y=0.981 and Q^2=0.762 in positive mode and R^2Y=0.999 and Q^2=0.531 in negative mode. NMIBC patients were clearly distinguished from MIBC. The models were confirmed by the 200 permutation test indicating that they were without overfitting (Supplemental Fig. 5) and can be used in the next step of the analysis. Then, VIP values based on the PLS-DA model were used to screen for differential metabolites. VIP >1 as one of the indicators for screening differential metabolites. AVOVA tests were then performed for data that conformed to normal distribution and homogeneity of variance. Welch AVOVA tests were performed for data that conformed to normal distribution and heterogeneity of variance, and used Dunnett T3 method to correct for multiple comparisons. Non-parametric tests (Kruskal-Wallis tests) were performed for data that did not conform to normal distribution using Dunn's test to correct. Finally, variables with VIP > 1 and adjusted p < 0.05 were defined as differential metabolites.

The differential metabolites obtained from the above screening were identified by MS-DIAL, HMDB and Pubchem. Based on these analyses, a total of eight differential metabolites were screened and identified to differentiate NMIBC from MIBC (Fig. 3, Supplemental Table 2).

3.4. ROC curve analysis and biomarker panel development

3.4.1. ROC analysis of potential biomarkers distinguishing between BC patients and healthy controls

ROC analyses were commonly used in metabolomics to assess the diagnostic performance of biomarkers. Area under the ROC curve (AUC) is usually utilised as indicators to assess the diagnostic ability of biomarkers. Biomarker with AUC > 0.7 is considered to have good diagnostic capability. In this study, 26 potential biomarkers were analysed by ROC to assess their ability to distinguish bladder cancer patients from healthy individuals (Table 2). Among them, nine potential biomarkers with AUC > 0.8 demonstrated promising diagnostic performance, which were urate, creatinine, 4-Androstene-3α,17β-diol, citric acid, 3-Indoxyl sulfate, aconitic acid, 5|A-Androst-1-ene-3|A,17|A-diol, *N*-Acetyl-*N*-formyl-5-methoxykynurenamine and taurine.

3.4.2. Biomarker panel development.

To further obtain a panel of differential metabolite with better diagnostic performance for the diagnosis of bladder cancer, we performed a binary logistic regression analysis and an optimized algorithm

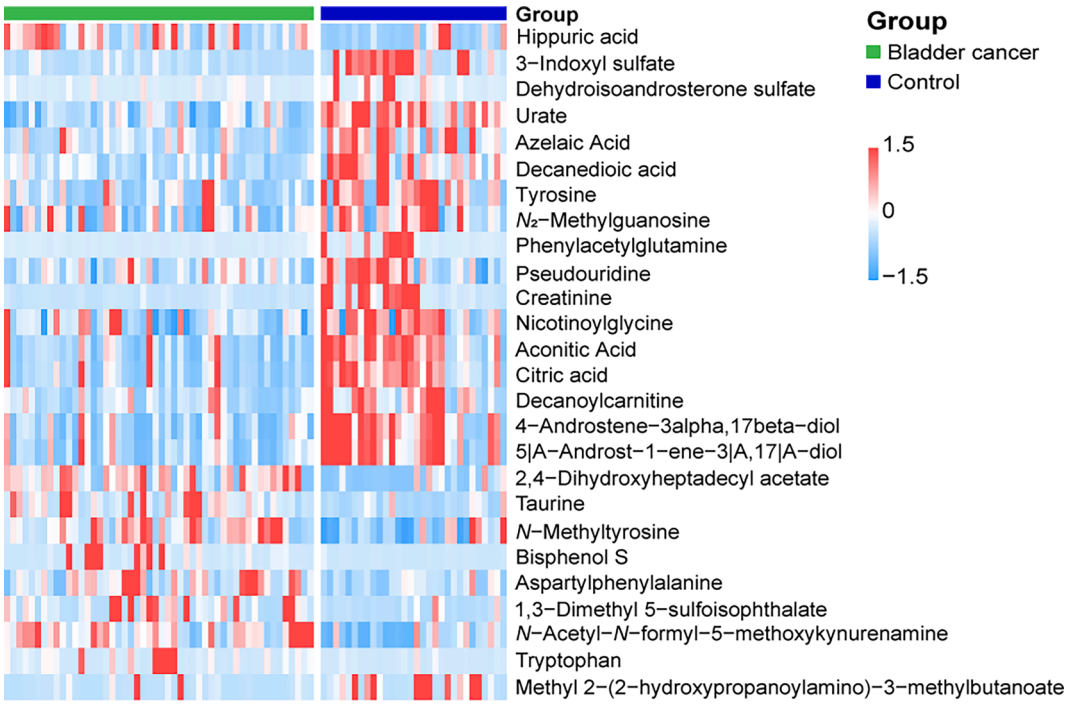


Fig. 2. Heatmap of changed metabolites between BC patients and healthy controls.

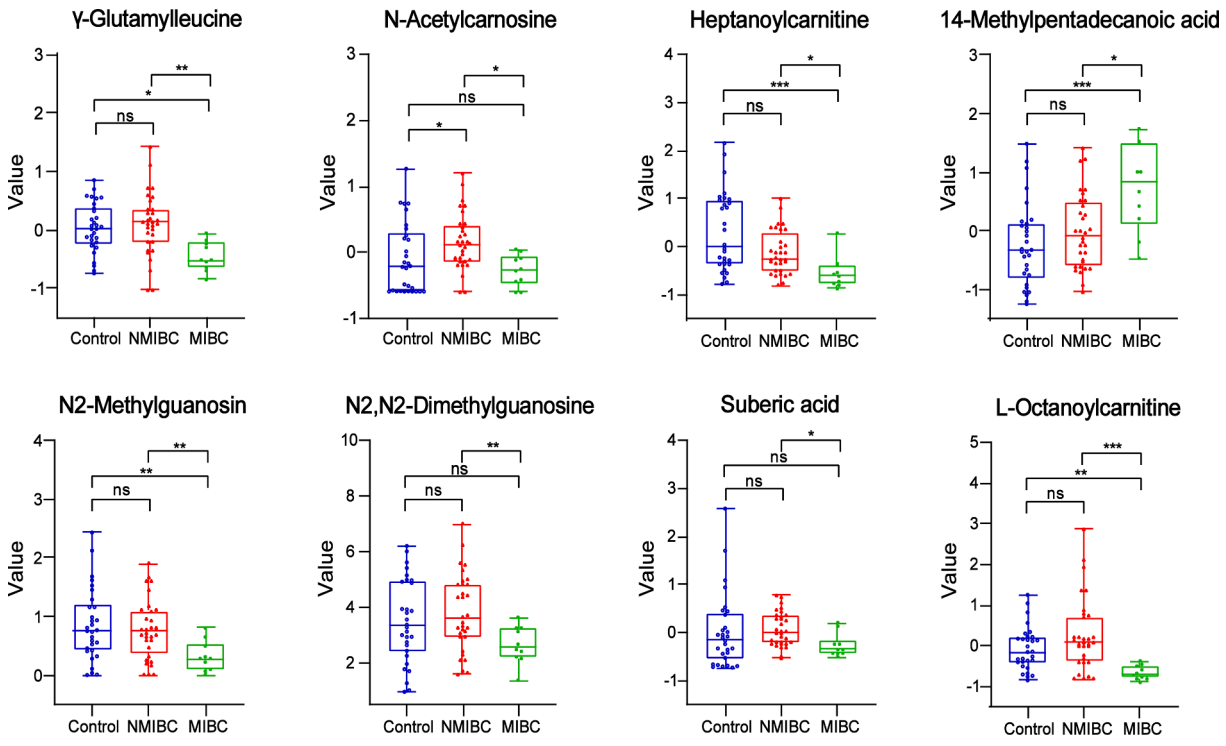


Fig. 3. Eight potential biomarkers for differentiating NMIBC patients from MIBC patients.

of the forward stepwise method for the 26 potential biomarkers mentioned above in order to construct a predictive model function for diagnosing bladder cancer. Finally, a biomarker panel containing 3-Indoxyl sulfate, urate and 4-Androstene-3α, 17β-diol was obtained, which could well distinguish the BC patients from healthy controls, with the predictive ability of 96.3% (Supplemental Table 3). The Omnibus test of the model coefficients satisfies $P < 0.05$, indicating that the model is significant (Supplemental Table 4). As Supplemental Table 5 and 6 show, Model quantitative evaluation and qualitative evaluation of goodness-of-fit effect are better. The resulting best model is:

Logit(P) = 3-Indoxyl sulfate × (−0.225) + Urate × (−7.053) + 4-Androstene-3alpha,17beta-diol × (−8.992) + 10.982

That is, as each unit of 3-Indoxyl sulfate increases, Logit(P) decreases by an average of 0.225 units. Each unit of urate increases, Logit(P) decreases by an average of 7.053 units. Each unit of 4-Androstene-3alpha,

Table 2
The ROC results of differential metabolites for Distinguishing BC patients from healthy controls.

No.	Compound	AUC	95% CI	Specificity (%)	Sensitivity (%)	Trend(BC vs Control)
1	Urate	0.915	0.843–0.965	80	80	↓####
2	Creatinine	0.862	0.758–0.943	80	80	↓####
3	4-Androstene-3α,17β-diol	0.846	0.753–0.926	80	80	↓####
4	3-Indoxyl sulfate	0.842	0.738–0.927	80	70	↓####
5	Citric acid	0.841	0.750–0.928	80	80	↓####
6	5|A-Androst-1-ene-3|A,17|A-diol	0.837	0.738–0.915	80	70	↓####
7	Aconitic Acid	0.829	0.732–0.916	90	80	↓####
8	*N*-Acetyl-*N*-formyl-5-methoxykynurenamine	0.806	0.683–0.891	80	80	↑####
9	Taurine	0.802	0.699–0.886	80	70	↑####
10	Decanedioic acid	0.784	0.649–0.89	80	70	↓####
11	2,4-Dihydroxyheptadecyl acetate	0.773	0.657–0.869	80	80	↑####
12	*N*-Methyltyrosine	0.770	0.647–0.878	80	70	↑###
13	Dehydroisoandrosterone sulfate	0.752	0.638–0.851	70	70	↓###
14	Decanoylcarnitine	0.748	0.628–0.859	70	70	↓###
15	Azelaic Acid	0.745	0.621–0.853	80	70	↓###
16	Hippuric acid	0.745	0.625–0.847	70	70	↑###
17	Tyrosine	0.729	0.592–0.843	80	70	↓###
18	Methyl 2-(2-hydroxypropanoylamino)-3-methylbutanoate	0.713	0.592–0.832	80	60	↓##
19	Nicotinoylglycine	0.702	0.575–0.836	80	70	↓##
20	1,3-Dimethyl 5-sulfoisophthalate	0.700	0.587–0.805	60	80	↑##
21	Aspartylphenylalanine	0.699	0.573–0.821	60	70	↑##
22	Bisphenol S	0.693	0.574–0.805	60	80	↑##
23	Phenylacetylglutamine	0.679	0.541–0.800	80	50	↓##
24	Pseudouridine	0.642	0.488–0.777	80	60	↓#
25	Tryptophan	0.640	0.523–0.759	50	80	↑#
26	N_2-Methylguanosine	0.623	0.495–0.742	60	60	↓#

The up arrows (or down arrows) represented the relative up-regulation (or down-regulation) of the differential lipids. #: $p < 0.05$, ##: $p < 0.01$, ###: $p < 0.001$, ####: $p < 0.0001$.

17beta-diol increases, Logit(P) decreases by an average of 8.992 units. Combined with OR, the higher the level of 3-Indoxyl sulfate, urate and 4-Androstene-3alpha, 17beta-diol, the lower the possibility of bladder cancer.

Then we investigated the predictive ability of the biomarker panel using AUC. The AUC value of the biomarker panel was 0.983, which was notably higher than that of any single metabolite (Fig. 4). In addition, the sensitivity of the biomarker panel was 0.9, the specificity was 1, indicating that it has a high accuracy in the diagnosis of bladder cancer. In order to improve the reliability and robustness of the results, internal validations were performed to validate the biomarker panel we constructed through cross-validation (K-fold cross-validation). The results show that the biomarker panel can accurately differentiate between bladder cancer patients and healthy individuals with 100% accuracy and AUC in the testing set. The results suggest that co-diagnosis with biomarker panel can better differentiate bladder cancer patients from healthy controls.

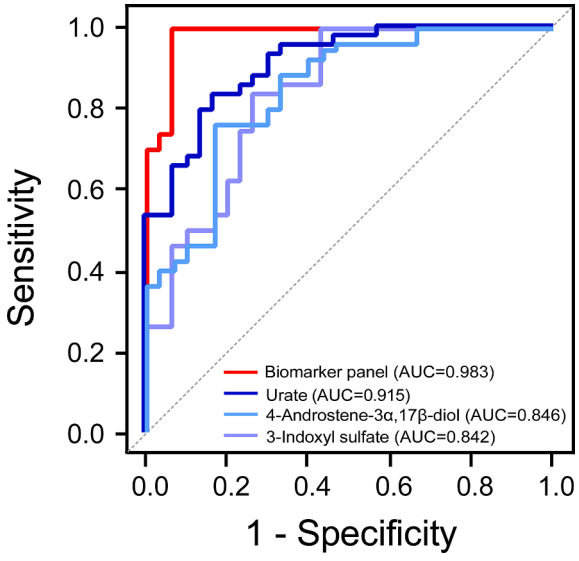


Fig. 4. ROC curves for metabolic biomarkers and biomarker panel to differentiate BC patients from healthy controls.

3.4.3. ROC analysis of potential biomarkers distinguishing between NMIBC patients and MIBC patients

ROC analyses were performed on the eight differential metabolites identified above, and the AUC of eight differential metabolites were > 0.7, indicating that all of them could well differentiate the NMIBC group from the MIBC group (Table 3). The top three metabolites with the AUC values (γ-Glutamylleucine, L-Octanoylcarnitine, Heptanoylcarnitine) were selected to form a biomarker panel for co-diagnosis. ROC analysis was then performed on the biomarker panel, the result showed that the AUC value of this biomarker panel was 0.941, the sensitivity and the specificity were both 0.9, which could better differentiate between NMIBC patients and MIBC patients (Fig. 5). Similarly, in order to improve the reliability and robustness of the results, internal validations were performed through cross-validation (K-fold cross-validation). The biomarker panel used to differentiate between NMIBC and MIBC had good diagnostic capability, with an accuracy and AUC of 87.5% and 91.7% in testing set, respectively.

3.5. Pathway analysis of differential metabolites

To investigate the effect of disease on differential metabolites, Kegg pathway analysis was performed using Metaboanalyst 5.0. According to the result, a total of twelve metabolic pathways are disrupted. The mainly impacted pathways, which $p < 0.05$ and impact value > 0.1, are Citrate cycle (TCA cycle), Tryptophan metabolism, Phenylalanine, tyrosine and tryptophan biosynthesis and Taurine and hypotaurine metabolism (Fig. 6). The detailed information of pathways analysis is shown in Supplemental Table 7. In addition, we constructed compound-reaction-enzyme-gene networks by MetScape. Supplemental Fig. 6 shows the genes associated with differential metabolites.

Table 3
The ROC results of differential metabolites for Distinguishing NMIBC patients from MIBC patients.

No.	Compound	AUC	95% CI	Specificity (%)	Sensitivity (%)	Adjusted P Value (NMIBC vs. MIBC)	Trend(NMIBC vs MIBC)
1	L-Octanoylcarnitine	0.884	0.762–0.970	90	80	3.0×10^{-4}	↓
2	γ-Glutamylleucine	0.850	0.711–0.945	80	80	4.3×10^{-3}	↓
3	Heptanoylcarnitine	0.819	0.620–0.953	80	70	2.8×10^{-2}	↓
4	*N*-Acetylcarnosine	0.816	0.664–0.934	80	70	2.8×10^{-2}	↓
5	Suberic acid	0.809	0.637–0.956	80	80	3.7×10^{-2}	↓
6	14-Methylpentadecanoic acid	0.781	0.600–0.925	80	70	1.1×10^{-2}	↑
7	N_2-Methylguanosin	0.781	0.622–0.905	80	70	2.3×10^{-3}	↓
8	N_2, N_2-Dimethylguanosine	0.753	0.601–0.877	90	60	3.6×10^{-3}	↓

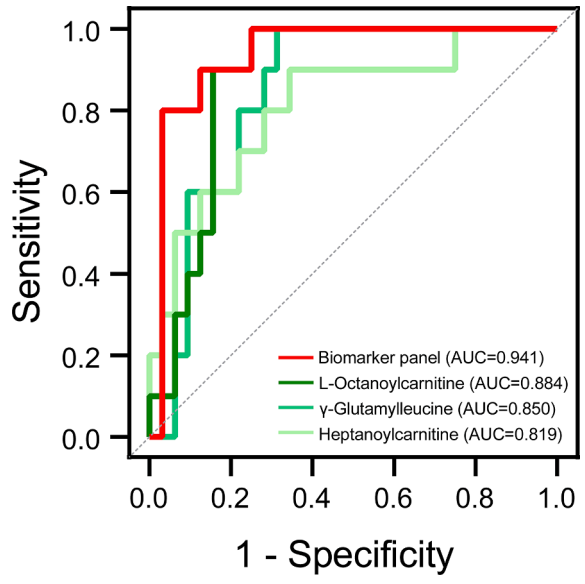


Fig. 5. ROC curves for metabolic biomarkers and biomarker panel to differentiate NMIBC patients from MIBC patients.

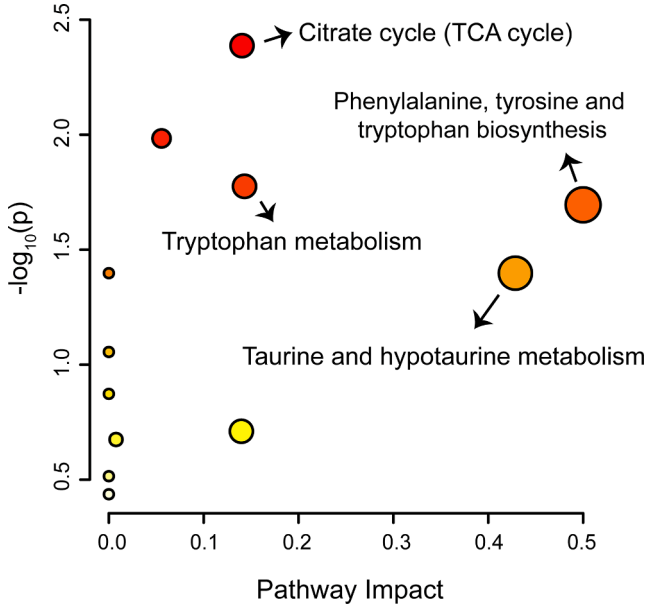


Fig. 6. Metabolic pathway analysis of 26 differential metabolites between BC patients and healthy controls. The P value (y-axis) was represented by the color of the circle and the pathway impact (x-axis) was indicated by the size of the circle. The involved pathways of the circles are indicated.

3.6. Correlation analysis

Among the eight differential metabolites that used to differentiate between NMIBC and MIBC, two differential metabolites were significantly correlated with BC stages (Fig. 7). Of these two metabolites, 14-Methylpentadecanoic acid (r=0.39) was positively associated with the development of bladder cancer, and heptanoylcarnitine (r=-0.43) was negatively associated with the development of bladder cancer.

In addition, to explore the potential impact of age and gender on differential metabolites, we explored the correlation between the level of differential metabolites, age and gender in bladder cancer patients (Supplemental Fig. 7). Among them, two androgens have higher levels in male, which are 4-Androstene-3α, 17β-diol (r=-0.33) and 5|A-Androst-1-ene-3|A,17|A-diol (r=-0.42). And five differential metabolites are negatively correlated with age, which are tyrosine (r=-0.32), Methyl 2-(2-hydroxypropanoylamino)-3-methylbutanoate (r=-0.43), 4-Androstene-3α, 17β-diol (r=-0.34), 5|A-Androst-1-ene-3|A,17|A-diol (r=-0.34) and N_2-Methylguanosine (r=-0.41). That is, levels of these five substances are negatively correlated with the development of bladder cancer.

4. Discussion

Metabolomics can be used to identify potential biomarkers for disease diagnosis and to understand the mechanisms by which diseases

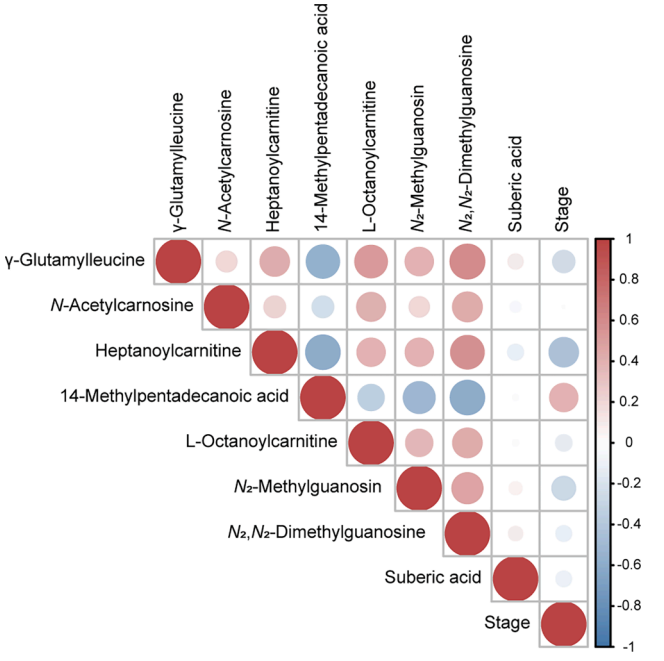


Fig. 7. Correlation heatmap illustrating the Pearson r calculated between stage and the levels of eight differential metabolites, |r| > 0.3 is considered to be relevant.

occur. In this study, we identified 26 differential metabolites in urine samples from 50 bladder cancer patients and 30 healthy controls using untargeted metabolomics. These 26 differential metabolites are mainly primarily associated with energy metabolism and amino acid metabolism in vivo. In addition, we also performed untargeted metabolomics analyses in 40 NMIBC patients and 10 MIBC patients out of 50 bladder cancer patients. A total of eight differential metabolites were ultimately identified, whose levels differed significantly between NMIBC patients and MIBC patients. These findings suggested that there were differences in metabolic profiles among the NMIBC, MIBC and healthy controls, which might be related to the occurrence of bladder cancer.

In the present study, citric acid levels were significantly reduced in bladder cancer patients and the results were the same as in previous studies [20–22]. Urinary citric acid is a normal component of urine and a major inhibitor of kidney stone formation [23]. Decreased levels of citric acid reflect an increase in the fatty acids required for the conversion of citric acid to β-oxidation, which is necessary to support rapid proliferation of cancer cells. In addition, levels of aconitic acid were also significantly lower in bladder cancer patients compared to healthy controls. Citric acid and aconitic acid are key intermediates in the TCA cycle, and the reduced levels of both suggest that the TCA cycle is impaired in bladder cancer patients, which may also be related to the energy required for β-oxidation to support cancer cell proliferation [13]. Moreover, reduction in citric acid levels in bladder cancer patients may be related to the Warburg effect [24].

Dysregulation of carnitine and several carnitine metabolite levels has been widely mentioned in studies related to bladder cancer [25,26]. It has been found in previous studies that carnitine metabolites, including 3-hydroxyoctanoyl carnitine, 2-octenoylcarnitine, 3-hydroxydecanoyl carnitine, octanoylcarnitine, 9-decenoylcarnitine, decanoylcarnitine, linolenyl carnitine, arachidylcarnitine, were higher in bladder cancer patients [13]. Huang et al. found that, in bladder patients, the level of Carnitine C8:1, Carnitine C9:1 and Carnitine C10:1 was lower, compared to the healthy controls [27]. In addition, there were differences in the levels of palmitoyl and octanoylcarnitine between high-grade and low-grade bladder cancers [28]. Carnitine and its metabolites play a key role in fatty acid metabolism. Fatty acids are one of the main sources of energy in the body, but they cannot access directly into the mitochondria for fatty acid oxidation. Carnitine acylates fatty acids to produce carnitine esters, which allow fatty acids to cross the inner mitochondrial membrane and enter the mitochondria, where they participate in β-oxidation reactions to produce energy [20]. In this study, the significant down-regulation of decanoylcarnitine levels in bladder cancer patients compared to healthy controls, which is the same as the results of previous studies [12], suggested dysregulation of fatty acid β-oxidation in bladder cancer patients. The levels of heptanoylcarnitine and L-Octanoylcarnitine were also significantly reduced in MIBC patients compared to NMIBC patients, further confirming the disordered of fatty acid transport and altered energy metabolism processes in bladder cancer patients. Additionally, azelaic acid is a saturated dicarboxylic acid produced by the oxidation of fatty acids, which inhibits mitochondrial oxidoreductases in the respiratory chain and enzymes related to DNA synthesis [15]. Azelaic acid has been shown to have antiproliferative and cytotoxic effects on tumour cells [29]. In our present study, the urine levels of azelaic acid significantly decreased in bladder cancer patients. The results above hint that the abnormal fatty acid metabolism may contribute to the occurrence of bladder cancer.

As we all know, amino acids play important roles in many biological processes such as protein synthesis, immune response, neuroendocrine regulation, and organ maturation [30]. In the present study, we found that the levels of tryptophan, taurine, and aspartylphenylalanine increase notably and the levels of creatinine, phenylacetylglutamine and γ-glutamylleucine decrease significantly in bladder cancer patients.

The relationship between tryptophan metabolism and bladder cancer has been extensively studied [31]. Some studies showed that tryptophan levels were significantly higher in bladder cancer patients compared to healthy controls, consistent with the results of our study [32,33]. Dysregulation of tryptophan metabolism affects multiple biological pathways and enzymes. Bladder cancer-associated cells upregulate the expression of enzymes such as indoleamine 2, 3-dioxygenase (IDO) or tryptophan 2, 3-dioxygenase (TDO), which exacerbate tryptophan degradation, leading to tryptophan depletion and immune evasion, thus reducing the availability of tryptophan for important cellular functions [34]. In addition, production of Kynurenic acid and other tryptophan metabolites can modulate T-cell immunity through activation of inhibitory regulatory T cells and activation of aryl hydrocarbon receptors, thereby promoting cancer cell survival [35]. Phenylacetylglutamine is a metabolite of phenylalanine. After phenylalanine is metabolized to phenylacetic acid in the body, it combines with glutamine to form phenylacetylglutamine, which is then excreted in the urine [36]. Phenylacetylglutamine is a biomarker for patients with urea cycle disorders [20]. In this study, phenylacetylglutamine levels were reduced in bladder cancer patients, and the change in phenylacetylglutamine levels suggests a dysregulation of either phenylalanine or glutamine, which may result from an increase in the energy required for the growth and proliferation of cancer cells.

Taurine is an amino acid that is converted from a sulfur-containing amino acid and also serves as a substrate for binding to bile acids. In the organism, taurine has the ability to regulate the osmotic pressure of the body, antioxidant and free radical scavenging effects, thus it can prevent cellular damage [37]. Consistent with the results of existing studies, taurine levels in the urine of bladder cancer patients were upregulated, speculating that taurine has a certain protective effect on the cells, which is conducive to the proliferation and metastasis of cancer cells [38].

Creatinine is an amino acid derivative that is produced from creatine. As a metabolic waste, creatinine is transferred to the kidneys through the bloodstream and excreted through urine after glomerular filtration [39]. The level of creatinine in the blood and urine are used as the indicators for evaluating renal function. In a metabolomics study of gastric cancer, gastric cancer patients had higher levels of creatine and lower levels of creatinine in urine compared to normal subjects [40]. Similarly, in this study, BC patients had significantly lower creatinine levels in urine. Up to 94% of creatine is present in muscle tissue and is converted to creatinine by a non-enzymatic reaction and excreted into the urine by the kidneys [39]. There is a significant correlation between creatine metabolism and muscle, and abnormalities in creatine metabolism implied that muscle of cancer patients may relative deterioration.

γ-Glutamylleucine is one of the substances involved in the γ-glutamyl cycle [41]. The γ-glutamyl cycle plays an important role in maintaining glutathione homeostasis. It has been shown that when the γ-glutamyl cycle is inhibited, the level of γ-glutamylleucine is relatively elevated and the intracellular glutathione level is reduced [42]. In this study, the level of γ-glutamylleucine was relatively reduced in MIBC patients compared to NMIBC patients, speculating that with the development of bladder cancer, the γ-glutamyl cycle is active, with a decrease in γ-glutamyl leucine and an increase in glutathione. Glutathione has multiple physiological functions and plays an important role in tumour cells [43]. Many studies have shown that elevated cellular levels of glutathione are required for tumourigenesis and proliferation, which is associated with the production of higher levels of reactive oxygen species in tumour cells [43].

In addition, there are a number of dipeptides whose levels are dysregulated. Aspartylphenylalanine is incomplete breakdown products of protein metabolism. Changes in amino acids and dipeptides in urine reflect high levels of cell division, cell death and protein degradation [37].

It is worth noting that Bisphenol S was significantly upregulated in the urine of bladder cancer patients in this study. Bisphenol S is an endocrine disruptor, which is commonly used in the production of plastics and can usually enter the body through the environment and food [44]. Bisphenol S has been shown to be associated with the

progression of bladder cancer [45]. Bisphenol S is primarily metabolised by the liver and excreted through urine, which can be stored in the bladder for several hours, thus the bladder will have a prolonged exposure to Bisphenol S. During this process, Bisphenol S can decrease the energy metabolism of urinary tract epithelial cells while increasing the energy metabolism of bladder cancer cells, which can lead to the development of bladder cancer [45].

Overall, this study identified 26 differential metabolites that could be used to differentiate bladder cancer from healthy controls and found 8 metabolites that differed between non-invasive and invasive bladder cancer. We further constructed biomarker panel for diagnosis and grading with AUC as high as 0.981 and 0.941, specificity and sensitivity greater than 0.9, which can accurately diagnose bladder cancer and differentiate its stages. In order to ensure the reliability of the results, we also performed internal validation. Nevertheless, our study has some shortcomings. Although we identified differential metabolites through mass spectrometry information from multiple databases, we were unable to use standards to further confirm our results due to resource and experimental constraints. In addition, despite the internal validation of the obtained results, we recognize that external validation is a crucial step to improve the reliability and robustness of research results and that internal validation is not a complete substitute for independent external validation. Therefore, in future studies, we will externally validate our results with independent cohorts and plan to probe with larger sample sizes in the future and further confirm our findings with actual standards or targeted metabolomics.

5. Conclusion

Metabolomics has been widely used to discover biomarkers for diagnosing diseases. In our research, we identified 26 potential biomarkers for distinguishing bladder cancer patients from healthy controls in urine samples by metabolomics. These metabolites are closely associated with energy metabolism and amino acid metabolism. Based on metabolic biomarkers, binary logistic regression was employed to form a biomarker panel, which can improve diagnosis ability. The AUC value of biomarker panel was as high as 0.983 by ROC analysis, indicating a good diagnostic ability. Meanwhile, metabolomics was used to identify potential biomarkers for NMIBC patients and MIBC patients. Eight potential biomarkers were identified for distinguishing NMIBC patients from MIBC patients. Likewise, biomarker panel which consisted of the top three biomarkers in terms of AUC, have a promising diagnostic ability. Therefore, we believe that metabolomics will have a vital role to play in the future diagnosis of bladder cancer.

CRediT authorship contribution statement

Xingyu Shi: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Wenbin Zheng:** Writing – review & editing, Methodology, Investigation. **Binhong He:** Methodology, Investigation. **Longhui Huang:** Investigation. **Qisheng Zhong:** Methodology. **Yunfan Yang:** Methodology. **Ting Zhou:** Supervision, Funding acquisition. **Yong Huang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Science Foundation of China (No. 22374051), and the Guangzhou Applied Basic Research Program (No.202201010109).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cca.2024.120022.

Data availability

Data will be made available on request.

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